

Exercise 2: Implementing Dependency Injection

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository" class="com.library.repository.BookRepository"/>

    <bean id="bookService" class="com.library.service.BookService">
        <property name="bookRepository" ref="bookRepository"/>
    </bean>

</beans>
```

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void display() {
        System.out.println("Book Repository is working");
    }

}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void showService() {
        System.out.println("Book Service is working");
        bookRepository.display();
    }

}
```

LibraryManagementApplication.java

```
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class LibraryManagementApplication {
```

```
public static void main(String[] args) {
```

```
    ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
```

```
    BookService service = context.getBean("bookService", BookService.class);
```

```
    service.showService();
```

```
}
```

```
}
```

